

HARDIK SOOD

AI/ML Engineer — Drone & Autonomous Systems | LLMs & RAG | Computer Vision | NVIDIA AI Stack

+91 6377279565 • hardiksood8@gmail.com • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

SUMMARY

AI/ML Engineer with hands-on experience across computer vision, edge-AI deployment, and large language model systems (RAG, fine-tuning, LLM-driven automation). Background spans real-time video analytics (YOLO, DeepStream, OpenCV), model optimization for edge/embedded inference (TensorRT, ONNX, Triton), and secure ML pipelines for cybersecurity applications. Now applying this computer-vision and edge-inference foundation to AI/ML for drone and autonomous-systems technology at Jarbits Pvt Ltd. 5 patents and 1 publication in applied AI/ML.

WORK EXPERIENCE

Senior AI/ML Engineer | Jarbits Pvt Ltd

June 2026 – Present • Kanpur, India

- *Drafted placeholder — replace with your real project details (see note below the résumé).*
- Working on AI/ML systems for drone and autonomous-systems technology, applying computer-vision and edge-inference experience (object detection/tracking, real-time video processing) from prior roles to aerial platforms.
- [Add specifics: e.g. onboard object detection/tracking model, flight-data pipeline, obstacle avoidance, autonomous navigation, swarm coordination, or telemetry analytics — name the model/stack and any measured outcome.]
- [Add specifics: hardware/software stack used — e.g. Jetson/edge device, flight controller (PX4/ArduPilot), ROS, gimbal/camera integration, etc.]

Assistant SDE – AI | C3iHub, IIT Kanpur

Nov 2025 – May 2026 • Kanpur, India

- Built an AI-driven L1 SOC analyst that automated triage and initial response for security alerts, reducing manual analyst workload.
- Integrated LLMs to turn complex security events into actionable, human-readable insights for SOC teams, and to auto-summarize detection-rule logic (including Suricata rule syntax) for faster analyst decision-making.
- Developed a hybrid ML model for PLC log anomaly detection, identifying targeted industrial attacks with 95% accuracy.
- Supported secure ML pipeline development and vulnerability analysis tooling across multidisciplinary security research initiatives; conducted domain analysis for customized threat-monitoring frameworks.

Apprentice, AI | CCS Computer Pvt. Ltd.

Feb 2025 – Oct 2025 • Delhi, India

- Fine-tuned a 7B-parameter Hugging Face model with LoRA/PEFT for efficient adaptation in resource-constrained environments.
- Built video analytics pipelines (OpenCV, GStreamer, NVIDIA DeepStream) covering object detection, face recognition, and real-time tracking; trained and deployed YOLOv5/YOLOv8 models integrated with DeepSORT re-identification.
- Implemented multilingual OCR (PaddleOCR) for scanned documents and video frames; built a handwritten text recognition system using NVIDIA NeMo encoder-decoder + CTC pipelines.
- Trained a RoBERTa model, converted it to ONNX, optimized with TensorRT, and deployed on NVIDIA Triton Inference Server for scalable low-latency inference.
- Built RAG knowledge-grounded QA workflows using NVIDIA NeMo and FAISS; developed a Node.js backend (JWT auth, MySQL) for a full-scale end-to-end project — recognized with a formal appreciation email from company leadership.

Intern | Vodafone Idea Foundation

Oct 2024 – Nov 2024 • India

- Analyzed agriculture and technology data, generating insights that contributed to a 15% efficiency increase for client processes; conducted case studies using advanced analytics and language-model-driven insights.

Data Science Summer Intern | Code Alpha

Jun 2024 – Jul 2024 • India

- Developed and deployed predictive models for customer behavior, achieving 92% accuracy on predictive tasks.

PUBLICATION

“Effective Techniques in News Text Summarization: An Evaluation of Extractive, Abstractive, and Hybrid Approaches,” GRENZE International Journal of Engineering and Technology, Vol. 11, Issue 1, pp. 4664–4689, 2025 — benchmarked transformer-based models for accuracy, linguistic coherence, and relevance in news summarization.

PATENTS

- Parking Management System (Jun 2025) — Smart ANPR & RFID-based parking automation.
- Automatic Audio Device Power Limiting System (Jun 2025) — MCU-based hearing-protection device.
- Biometric Safety Pendant (Feb 2025) — Wearable with fingerprint + GPS for emergency response.
- Multi-Color Marker with Ink Level Detection (Feb 2025) — LED-based low-ink alert system.
- Multi-Level Marker (Feb 2025) — Marker design with integrated ink monitoring.

SELECTED PROJECTS

- Product Image Intelligence Platform — Multi-model LLM app that generates product titles, descriptions, and feature lists from images, with Google Shopping lookups via SerpAPI.
- AI-Driven RAG Document Assistant — ChatGPT-style Q&A over PDFs (Python/Streamlit) using FAISS retrieval and NVIDIA LLaMA-based LLMs, with hallucination-gating so answers are limited to source content.
- Real-Time Vehicle Monitoring System — Vehicle detection, speed estimation, and automated license-plate recognition on live video for transportation analytics.
- Medical Document OCR & Summarization — Multilingual OCR and summarization pipeline generating per-patient summaries from medical documents/images.
- The Active Learning Platform — NVIDIA AI Foundation Models-powered curriculum generator with a hallucination-free resource-mapping layer and tiered practice arena; multilingual (Hindi/English) UI.
- Agricultural Yield Prediction Tool — ML pipeline forecasting crop yields to support farmer/agribusiness decisions.

TECHNICAL SKILLS

- Computer Vision / Edge AI: YOLOv5/v8, DeepSORT, OpenCV, GStreamer, NVIDIA DeepStream, PaddleOCR
- LLMs & RAG: Hugging Face Transformers, LoRA/PEFT fine-tuning, FAISS, NVIDIA NeMo & NIM, prompt/agent pipelines
- Model Optimization & Serving: ONNX, NVIDIA TensorRT, NVIDIA Triton Inference Server
- Programming: Python, R, SQL, C++, HTML, CSS
- Cloud & MLOps: AWS, Azure, Google Cloud, Docker, Flask, Gradio
- Data & Visualization: MySQL, MongoDB, Hive, Power BI, Tableau, Excel

EDUCATION

MSc, Data Science | Chandigarh University *Jul 2023 – Jul 2025 • GPA: 7.35*

BCA, Computer Science | Maharaja Ganga Singh University, Bikaner *Apr 2019 – Apr 2022 • GPA: 7.1*

CERTIFICATIONS

- Oracle Cloud Infrastructure Certified Generative AI Professional (Sep 2025)
- Develop Generative AI Apps in Azure — Microsoft (Jun 2025)
- Google Data Analytics Professional Certificate • Natural Language Processing on Google Cloud
- Tata Data Visualization: Empowering Business with Effective Insights (Jun 2024)
- Accenture North America Data Analytics and Visualization (Jul 2024)
- Python Certification • Java Basics Certification